Dataset Report — Sentinel-2 Vegetation Indices

Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France

SENTINEL-2 OPTICAL

Dataset role: Parcel-level vegetation indices time series (NDVI, EVI, NDWI)
Provider: Copernicus Sentinel-2 mission (ESA) — via Google Earth Engine
Asset: COPERNICUS/S2_SR_HARMONIZED · **Resolution:** 10–20 m
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1. Source and provenance

1.1 Sentinel-2 mission

Sentinel-2 is a constellation of two identical satellites (S2A launched 2015, S2B launched 2017), operated by the European Space Agency under the Copernicus programme. Together they revisit each location every ~5 days at the equator, and ~2-3 days at mid-latitudes such as France. The MultiSpectral Instrument (MSI) records 13 spectral bands at 10 m, 20 m and 60 m resolution.

1.2 Why Sentinel-2 for this project

Sentinel-2 is the natural choice for parcel-level wheat monitoring in Europe because:

- Free, open data with global coverage
- 10 m resolution captures parcel-scale heterogeneity
- NIR + Red-Edge + SWIR bands enable derivation of NDVI, EVI, NDWI
- 5-day revisit allows detailed phenological tracking
- Harmonized SR product (L2A) provides analysis-ready surface reflectance

1.3 Access method: Google Earth Engine

Sentinel-2 imagery cannot be processed scene-by-scene locally — each Level-2A tile is ~1 GB and ingestion requires substantial compute. Google Earth Engine (GEE) provides a planet-scale processing platform with the Sentinel-2 archive pre-ingested, accessible via JavaScript or Python APIs. All extraction was performed server-side on GEE.

2. Acquisition workflow

Step 1 — Asset upload. The 1,500-parcel CSV (lat/lon centroids) was uploaded to GEE as a FeatureCollection asset (`parcelles_ble_cvl_sample`, EPSG:4326). This asset is reused across both Sentinel-2 and Sentinel-1 pipelines.

Step 2 — Collection filtering. The `COPERNICUS/S2_SR_HARMONIZED` collection was filtered by: date range (2019-10-01 → 2024-07-31), bounding region (parcels), and scene-level cloud cover (`CLOUDY_PIXEL_PERCENTAGE < 60 %`). This yielded 3,862 candidate scenes.

Step 3 — Cloud masking. For each scene, the Scene Classification Layer (SCL) was used to mask classes 3 (cloud shadow), 7, 8, 9 (cloud probabilities), 10 (cirrus), and 11 (snow). Surface reflectance was rescaled by dividing by 10,000 to obtain values in [0, 1].

Step 4 — Vegetation index computation. Three indices were computed per scene: NDVI from B8/B4, EVI from B8/B4/B2 (gain 2.5, coefficients 6 and -7.5), and NDWI from B8/B11. Computation is lazy in GEE — indices are calculated only when sampled.

Step 5 — Composite construction (15j and 7j). For each fixed-length window, 5 statistics (median, mean, max, min, stdDev) of NDVI/EVI/NDWI were computed across all valid scenes. Empty windows (no cloud-free scenes) produced 15 NaN bands.

Step 6 — Parcel-level extraction. For each composite or scene, `reduceRegions(parcels, mean, scale=20)` was applied to obtain the parcel-mean value of each band.

Step 7 — Export to Google Drive. Seven CSV files were exported via `Export.table.toDrive()`: one 15-day composite, one 7-day composite, and five raw-scene files (one per wheat campaign year).

3. Files produced

File	Size	Use
`sentinel2_composites_15days_cvl.csv`	42 MB	Classical ML (RF, XGBoost) — fast loading, low NaN rate
`sentinel2_composites_7days_cvl.csv`	~85 MB	Sequence models (LSTM, GRU) — finer resolution
`sentinel2_raw_{2020-2024}.csv`	~40 MB × 5	Maximum flexibility — any custom aggregation in Python

4. Key methodological decisions

Decision: use GEE rather than ESA's Sentinel Hub or local processing.
Rationale: GEE provides the full Sentinel-2 archive pre-ingested, with cloud-masking and band arithmetic at planet-scale. Local processing of ~3,900 scenes (~4 TB raw) is infeasible.

Decision: use the harmonized SR collection (`S2_SR_HARMONIZED`) rather than the older `S2_SR`.
Rationale: the harmonized version corrects the 2022 processing baseline change (Baseline 04.00) that shifted surface reflectance values by ~1000 DN. Using it ensures the 2020-2024 series is comparable without further correction.

Decision: SCL-based cloud masking rather than QA60 bitmask.
Rationale: SCL (Scene Classification Layer) is the modern, supervised classification provided by ESA at L2A. It is more accurate than the legacy QA60 cloud bitmask and is the recommended approach in recent literature.

Decision: three temporal granularities exported (15j, 7j, raw).
Rationale: different ML architectures benefit from different temporal resolutions. Tree-based models work well with wide-format 15-day data; sequence models like LSTM benefit from finer 7-day resolution; Transformers and custom architectures may need raw scenes. Providing all three avoids re-running GEE for each architecture.

Decision: raw export split into one CSV per harvest year.
Rationale: a single file would exceed 5 million rows (GEE export limit). Per-year splits keep each file at ~1.15 M rows, comfortably below the limit.

Decision: 5 statistics (median, mean, max, min, stdDev) per composite, not just median.
Rationale: capturing dispersion (stdDev) and extreme values (max, min) in each window provides ML models with information about within-window stability and outlier behaviour. This costs marginal file size but maximises feature richness.

5. Initial robustness issue and fix

The first 7-day composite export failed with the error `Image.reduceRegions: Image has no bands` at window 17 (winter 2020). Investigation revealed that some 7-day windows were entirely cloud-covered, producing a composite with no bands.

The fix uses `ee.Algorithms.If` in the composite function: if a window contains zero valid scenes, a ghost image with 15 masked bands is returned instead of an empty image. The downstream sampling function then filters composites with `n_images > 0` to skip empty windows cleanly.

Lesson learned: SAR/optical pipelines on GEE must always guard against empty collections at every reduction step. `ee.Algorithms.If` with a properly-shaped fallback image is the standard idiom.

6. Data quality summary

6.1 Coverage

Aspect	15-day composite	7-day composite	Raw (per year)
Total rows	177,000	~380,000	~1,120,000 each
Valid (non-NaN)	74,2 %	~65 % (est.)	~7 % (structural)*
Critical period (May–Jul)	~80–90 % valid	~75–85 % valid	~10 scenes/parcel/month

** Raw rows include all (parcel × scene_date) combinations; rows are NaN when the parcel is outside the scene footprint. This is structural, not data loss.*

6.2 Plausibility checks performed

- **NDVI range:** all values within [-1, 1] as expected.
- **Phenological profile:** canonical wheat curve confirmed (peak ~0,80 in April–May, collapse in July).
- **Inter-annual stability:** 5 seasons follow nearly identical patterns when aligned on DOY.
- **EVI outliers:** 159 rows (0,12 %) flagged with `|EVI| > 1,5` — numerical artifacts; filter recommended.
- **NDVI/NDWI correlation:** r = 0,93 — informational redundancy noted.

7. Deliverables

Item	Description
7 CSV files	15j + 7j composites, 5 raw yearly files
Kaggle Dataset	`sentinel2-indices-cvl-1500-wheat-parcels`
Data Dictionary	`Data_Dictionary_Sentinel2.pdf`
EDA Report	`EDA_Report_Sentinel2_15days.pdf` (9 pages, English) + FR version
GEE script	Reproducible JavaScript pipeline (saved in user's GEE repository)

8. Relevant observations

- **Seasonal coverage gap:** winter months (Dec–Feb) show ~40-50 % NaN due to persistent cloud cover. This is the primary motivation for adding Sentinel-1 SAR (all-weather complement).
- **Recommended pre-processing:** filter `|EVI| < 1,5`, optionally apply temporal interpolation using the raw dataset to fill winter gaps.
- **Feature engineering priority:** derived features (peak_doy, senescence_doy, NDVI integral) likely carry most of the predictive signal — see EDA report.
- **Reproducibility:** the GEE script is parameterised by the parcels asset and date range. Extending to other regions or crops requires only updating these two inputs.